**Supplementary file 6- Articles related to uncertainty classifications developed from reviewing the literature**

| Study | Context | Classification of uncertainties |
|---|---|---|
| **Fox 1957**[7] | Uncertainties experienced by doctors | • Uncertainty in knowledge which can never be mastered<br>• Uncertainty of reality<br>• Uncertainty of knowledge or imperfect mastery. |
| **Light 1979**[8] | Uncertainties experienced by medical residents (trainees) | • Uncertainties arising from relationship of students with their instructors<br>• Uncertainties from diagnosis, analysis, and treatment of those problems and from the clients or patients' responses. |
| **Mishel 1988**[15] | Uncertainties in nursing and related fields | • Ambiguity<br>• Complexity<br>• Deficient information<br>• Unpredictability of patients' course of illness and outcomes |
| **Smithson 1989**[17] | Uncertainties in healthcare in general | Lack of knowledge which arises from<br>• Probabilities (likelihood)<br>• Ambiguity (multiple causes)<br>• Vagueness (fuzzy boundaries) |
| **Babrow 1998**[30] | Uncertainty experienced by physicians and by extension patients | • Complexity<br>• Qualities of information<br>• Probability<br>• Structure of information and lay epistemology |
| **Penrod 2001**[19] | Uncertainties related to patients described in the field of nursing | Experience of patients with illness regarding<br>• Prognosis<br>• Course of treatment<br>• Choice of provider<br>• Family relationships. |
| **Begun 2004**[20] | Uncertainties in the organisational environment of healthcare system | Environmental uncertainty<br>• Complexity<br>• Dynamism |
| **Han 2011**[2] | Uncertainties in healthcare | • Source<br>• Issue<br>• Loci |

| Study | Context | Classification of uncertainties |
|---|---|---|
| **Djulbergovic 2011[22]** | Uncertainties in clinical practice and research | • Epistemic<br>• Subjective<br>• Linguistic uncertainty<br>• Probability |
| **Wray 2015[27]** | Uncertainties experienced by physicians | • Informational due to knowledge deficits<br>• Intrinsic uncertainties inherent in daily clinical care |
| **Hong 2020[12]** | Uncertainties in communicating cancer related genetic risk information to patients | Uncertainty about understanding genetic information were classified as<br>• Future cancer risks<br>• managing genetic information or mutation status<br>• Utility of genetic information<br>• Genetic test results before undergoing testing or receiving the results<br>• Impact of genetic results on family and life |
| **Lee 2020[11]** | Uncertainties pertaining to medical students | • Sources<br>• Subjective influences<br>• Responses to uncertainty |
| **Pomare 2019[5]** | Uncertainties in complex healthcare systems | Addition of additional issues to Han's framework<br>• Ethics<br>• Structures of care |